

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                        |
|----------------------|------------------------|
| United States Patent | 12398464               |
| Kind Code            | B2                     |
| Date of Patent       | August 26, 2025        |
| Inventor(s)          | Gomm; Troy Alan et al. |

---

### Substrate pedestal including backside gas-delivery tube

---

#### Abstract

A semiconductor substrate processing apparatus includes a vacuum chamber having a processing zone in which a semiconductor substrate may be processed, a process gas source in fluid communication with the vacuum chamber for supplying a process gas into the vacuum chamber, a showerhead module through which process gas from the process gas source is supplied to the processing zone of the vacuum chamber, and a substrate pedestal module. The substrate pedestal module includes a platen made of ceramic material having an upper surface configured to support a semiconductor substrate thereon during processing, a stem made of ceramic material having an upper stem flange that supports the platen, and a backside gas tube made of ceramic material that is located in an interior of the stem. The backside gas tube includes an upper gas tube flange that is located between a lower surface of the platen and an upper surface of the upper stem flange wherein the backside gas tube is in fluid communication with at least one backside gas passage of the platen and the backside gas tube is configured to supply a backside gas to a region below a lower surface of a semiconductor substrate that is to be supported on the upper surface of the platen during processing.

---

**Inventors:** Gomm; Troy Alan (Sherwood, OR), Linebarger, Jr.; Nick Ray (Beaverton, OR)

**Applicant:** Lam Research Corporation (Fremont, CA)

**Family ID:** 1000008777974

**Assignee:** Lam Research Corporation (Fremont, CA)

**Appl. No.:** 18/123153

**Filed:** March 17, 2023

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20230220549 A1   | Jul. 13, 2023    |

## Related U.S. Application Data

continuation parent-doc US 15662553 20170728 US 10655225 child-doc US 16863835  
division parent-doc US 16863835 20200430 US 11634817 child-doc US 18123153  
division parent-doc US 14710132 20150512 US 9738975 child-doc US 15662553

---

## Publication Classification

**Int. Cl.:** C23C16/458 (20060101); B32B37/10 (20060101); C23C16/505 (20060101);  
C23C16/52 (20060101); H01J37/32 (20060101); H01L21/67 (20060101); H01L21/683  
(20060101); H01L21/687 (20060101)

### U.S. Cl.:

CPC C23C16/4586 (20130101); B32B37/1018 (20130101); C23C16/4581 (20130101);  
C23C16/505 (20130101); C23C16/52 (20130101); H01J37/32577 (20130101);  
H01L21/67103 (20130101); H01L21/6831 (20130101); H01L21/68757 (20130101);  
H01L21/68785 (20130101); H01L21/68792 (20130101); B32B2457/00 (20130101)

## Field of Classification Search

**CPC:** C23C (16/4586); C23C (16/4581); C23C (16/505); C23C (16/52); H01L (21/67103);  
H01L (21/6831); H01L (21/68757); H01L (21/68785); H01L (21/68792); B32B  
(2457/00); B32B (37/1018); H01J (37/32577)

**USPC:** 118/728-732

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No. | Issued Date | Patentee Name      | U.S. Cl. | CPC        |
|------------|-------------|--------------------|----------|------------|
| 5569356    | 12/1995     | Lenz et al.        | N/A      | N/A        |
| 5688331    | 12/1996     | Aruga et al.       | N/A      | N/A        |
| 6035101    | 12/1999     | Sajoto et al.      | N/A      | N/A        |
| 6103074    | 12/1999     | Khominich          | N/A      | N/A        |
| 6149727    | 12/1999     | Yoshioka           | 269/21   | B05C 11/08 |
| 6223447    | 12/2000     | Yudovsky et al.    | N/A      | N/A        |
| 6464795    | 12/2001     | Sherstinsky et al. | N/A      | N/A        |
| 6494955    | 12/2001     | Lei et al.         | N/A      | N/A        |
| 6524428    | 12/2002     | Tamura et al.      | N/A      | N/A        |
| 6616767    | 12/2002     | Zhao et al.        | N/A      | N/A        |
| 6860965    | 12/2004     | Stevens            | N/A      | N/A        |
| 6962348    | 12/2004     | Fink               | N/A      | N/A        |
| 6997993    | 12/2005     | Yamaguchi et al.   | N/A      | N/A        |
| 7126808    | 12/2005     | Koo et al.         | N/A      | N/A        |
| 7241346    | 12/2006     | Hanamachi et al.   | N/A      | N/A        |
| 7274006    | 12/2006     | Okajima et al.     | N/A      | N/A        |
| 7632356    | 12/2008     | Tomita et al.      | N/A      | N/A        |
| 7645341    | 12/2009     | Kennedy et al.     | N/A      | N/A        |

|              |         |                    |            |                  |
|--------------|---------|--------------------|------------|------------------|
| 7737035      | 12/2009 | Lind et al.        | N/A        | N/A              |
| 7869184      | 12/2010 | Steger             | N/A        | N/A              |
| 8294069      | 12/2011 | Goto et al.        | N/A        | N/A              |
| 8394199      | 12/2012 | Tomita et al.      | N/A        | N/A              |
| 8801950      | 12/2013 | Lee                | N/A        | N/A              |
| 8840754      | 12/2013 | Hao                | N/A        | N/A              |
| 8851463      | 12/2013 | Angelov et al.     | N/A        | N/A              |
| 9490150      | 12/2015 | Tzu et al.         | N/A        | N/A              |
| 9728437      | 12/2016 | Tran               | N/A        | H01L<br>21/67103 |
| 9738975      | 12/2016 | Gomm               | N/A        | C23C 16/505      |
| 10147610     | 12/2017 | Lingampalli et al. | N/A        | N/A              |
| 10177024     | 12/2018 | Gomm et al.        | N/A        | N/A              |
| 10655225     | 12/2019 | Gomm et al.        | N/A        | N/A              |
| 11634817     | 12/2022 | Gomm et al.        | N/A        | N/A              |
| 2001/0042744 | 12/2000 | Tachikawa          | 219/544    | H05B 3/68        |
| 2002/0050246 | 12/2001 | Parkhe             | 118/500    | H01L<br>21/67109 |
| 2003/0136520 | 12/2002 | Yudovsky           | 118/723E   | C23C<br>16/4586  |
| 2003/0150563 | 12/2002 | Kuibara            | 118/728    | C04B 35/581      |
| 2003/0183341 | 12/2002 | Yamaguchi et al.   | N/A        | N/A              |
| 2004/0194710 | 12/2003 | Hanamachi          | 118/728    | C23C<br>16/4583  |
| 2006/0102595 | 12/2005 | Inoue              | N/A        | N/A              |
| 2006/0186109 | 12/2005 | Goto et al.        | N/A        | N/A              |
| 2006/0231388 | 12/2005 | Mullapudi et al.   | N/A        | N/A              |
| 2007/0040265 | 12/2006 | Umotoy             | 257/704    | C23C<br>16/45521 |
| 2007/0151517 | 12/2006 | Baik et al.        | N/A        | N/A              |
| 2007/0169703 | 12/2006 | Elliot et al.      | N/A        | N/A              |
| 2008/0078426 | 12/2007 | Miya et al.        | N/A        | N/A              |
| 2009/0235866 | 12/2008 | Kataigi            | 118/725    | C23C<br>16/4586  |
| 2009/0277388 | 12/2008 | Lubomirsky         | 118/725    | C23C<br>16/4586  |
| 2010/0163183 | 12/2009 | Tanaka et al.      | N/A        | N/A              |
| 2011/0064878 | 12/2010 | Suzuki et al.      | N/A        | N/A              |
| 2011/0096461 | 12/2010 | Yoshikawa et al.   | N/A        | N/A              |
| 2011/0206843 | 12/2010 | Gurary et al.      | N/A        | N/A              |
| 2011/0303154 | 12/2010 | Kim et al.         | N/A        | N/A              |
| 2012/0018416 | 12/2011 | Goto               | N/A        | N/A              |
| 2012/0031338 | 12/2011 | Lee                | N/A        | N/A              |
| 2012/0204796 | 12/2011 | Ruda Y Witt        | 118/728    | C23C<br>16/4583  |
| 2012/0211933 | 12/2011 | Goto               | N/A        | N/A              |
| 2013/0092086 | 12/2012 | Keil et al.        | N/A        | N/A              |
| 2014/0008349 | 12/2013 | Tzu                | 219/448.18 | H01L<br>21/67103 |
| 2014/0087587 | 12/2013 | Lind               | N/A        | N/A              |

|              |         |                        |          |                  |
|--------------|---------|------------------------|----------|------------------|
| 2014/0177123 | 12/2013 | Thach                  | 361/234  | H01L<br>21/67103 |
| 2014/0231019 | 12/2013 | Kajihara               | N/A      | N/A              |
| 2014/0302256 | 12/2013 | Chen et al.            | N/A      | N/A              |
| 2015/0011096 | 12/2014 | Chandrasekharan et al. | N/A      | N/A              |
| 2015/0056787 | 12/2014 | Brenninger             | N/A      | N/A              |
| 2015/0138687 | 12/2014 | Boyd, Jr.              | 361/234  | H01L<br>21/6833  |
| 2015/0376780 | 12/2014 | Khaja et al.           | N/A      | N/A              |
| 2016/0002779 | 12/2015 | Lin et al.             | N/A      | N/A              |
| 2016/0099164 | 12/2015 | Kataigi                | 228/49.5 | H01L<br>21/6833  |
| 2016/0225651 | 12/2015 | Tran                   | N/A      | H01L<br>21/67103 |
| 2016/0333475 | 12/2015 | Gomm et al.            | N/A      | N/A              |
| 2017/0321324 | 12/2016 | Gomm et al.            | N/A      | N/A              |
| 2020/0325578 | 12/2019 | Gomm et al.            | N/A      | N/A              |

## FOREIGN PATENT DOCUMENTS

| Patent No.  | Application Date | Country | CPC |
|-------------|------------------|---------|-----|
| 1461288     | 12/2002          | CN      | N/A |
| 1779940     | 12/2005          | CN      | N/A |
| 101154560   | 12/2007          | CN      | N/A |
| 106148915   | 12/2015          | CN      | N/A |
| 106148915   | 12/2019          | CN      | N/A |
| 112063995   | 12/2019          | CN      | N/A |
| 07153706    | 12/1994          | JP      | N/A |
| 2002093894  | 12/2001          | JP      | N/A |
| 2002110774  | 12/2001          | JP      | N/A |
| 2004152865  | 12/2003          | JP      | N/A |
| 2006128203  | 12/2005          | JP      | N/A |
| 2007165322  | 12/2006          | JP      | N/A |
| 2009054871  | 12/2008          | JP      | N/A |
| 2009256789  | 12/2008          | JP      | N/A |
| 2010109316  | 12/2009          | JP      | N/A |
| 2012182221  | 12/2011          | JP      | N/A |
| 2014523143  | 12/2013          | JP      | N/A |
| 2016213456  | 12/2015          | JP      | N/A |
| 2021061422  | 12/2020          | JP      | N/A |
| 20160133373 | 12/2015          | KR      | N/A |

## OTHER PUBLICATIONS

“Korean Application Serial No. 10-2016-0057390, Notice of Preliminary Rejection mailed May 28, 2023”, W English Translation, 4 pgs. cited by applicant

“U.S. Appl. No. 14/710,132, Decision on Pre-Appeal Brief Request mailed May 18, 2017”, 1 pg. cited by applicant

“U.S. Appl. No. 14/710,132, Final Office Action mailed Feb. 6, 2017”, 22 pgs. cited by applicant

“U.S. Appl. No. 14/710,132, Non Final Office Action mailed Sep. 9, 2016”, 14 pgs. cited by

applicant

“U.S. Appl. No. 14/710,132, Notice of Allowance mailed May 18, 2017”, 10 pgs. cited by applicant

“U.S. Appl. No. 14/710,132, Pre-Appeal Brief Request filed Mar. 17, 2017”, 5 pgs. cited by applicant

“U.S. Appl. No. 14/710,132, Response filed Jun. 3, 2016 to Restriction Requirement mailed Apr. 19, 2016”, 1 pg. cited by applicant

“U.S. Appl. No. 14/710,132, Response filed Nov. 29, 2016 to Non Final Office Action mailed Sep. 9, 2016”, 15 pgs. cited by applicant

“U.S. Appl. No. 14/710,132, Restriction Requirement mailed Apr. 19, 2016”, 5 pgs. cited by applicant

“U.S. Appl. No. 15/662,553, Non Final Office Action mailed Jun. 13, 2019”, 16 pgs. cited by applicant

“U.S. Appl. No. 15/662,553, Notice of Allowance mailed Feb. 3, 2020”, 10 pgs. cited by applicant

“U.S. Appl. No. 15/662,553, Response filed Dec. 11, 2019 to Non Final Office Action mailed Jun. 13, 2019”, 12 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Advisory Action mailed Feb. 25, 2022”, 4 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Corrected Notice of Allowability mailed Mar. 8, 2023”, 8 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Final Office Action mailed Dec. 6, 2021”, 21 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Non Final Office Action mailed Jul. 21, 2021”, 20 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Non Final Office Action mailed Aug. 1, 2022”, 18 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Notice of Allowance mailed Dec. 12, 2022”, 11 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Preliminary Amendment filed Jul. 23, 2020”, 12 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Response filed Feb. 7, 2022 to Final Office Action mailed Dec. 6, 2021”, 16 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Response filed Mar. 7, 2022 to Advisory Action mailed Feb. 25, 2022”, 17 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Response filed Jun. 28, 2021 to Restriction Requirement mailed Apr. 29, 2021”, 7 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Response filed Oct. 21, 2021 to Non Final Office Action mailed Jul. 21, 2021”, 16 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Response filed Nov. 1, 2022 to Non Final Office Action mailed Aug. 1, 2022”, 12 pgs. cited by applicant

“U.S. Appl. No. 16/863,835, Restriction Requirement mailed Apr. 29, 2021”, 7 pgs. cited by applicant

“Chinese Application Serial No. 201610312750.1, Office Action mailed Jun. 3, 2019”, W/ English Translation, 15 pgs. cited by applicant

“Chinese Application Serial No. 201610312750.1, Response filed Dec. 18, 2019 to Office Action mailed Jun. 3, 2019”, Amended Claims in English, 6 pgs. cited by applicant

“Chinese Application Serial No. 201610312750.1, Response to Examiner Interview filed Apr. 22, 2020”, English Translation of Amended Claims, 6 pgs. cited by applicant

“Chinese Application Serial No. 202010719108.1, Office Action mailed May 27, 2022”, w/ English translation, 17 pgs. cited by applicant

“Chinese Application Serial No. 202010719108.1, Office Action mailed Nov. 25, 2022”, w/o English Translation, 6 pgs. cited by applicant

“Chinese Application Serial No. 202010719108.1, Response filed Oct. 11, 2022 to Office Action

mailed May 27, 2022”, w/ English claims, 4 pgs. cited by applicant

“Japanese Application Serial No. 2016-92266, Notification of Reasons for Refusal mailed Aug. 11, 2020”, w/ English translation, 6 pgs. cited by applicant

“Japanese Application Serial No. 2016-92266, Response filed Nov. 6, 2020 to Notification of Reasons for Refusal mailed Aug. 11, 2020”, w/ English Claims, 10 pgs. cited by applicant

“Japanese Application Serial No. 2016-92266, Voluntary Amendment filed Jul. 12, 2019”, w/ English Claims, 11 pgs. cited by applicant

“Japanese Application Serial No. 2020-213139, Notification of Reasons for Refusal mailed Mar. 15, 2022”, w/ English Translation, 6 pgs. cited by applicant

“Japanese Application Serial No. 2020-213139, Response filed Sep. 8, 2022 to Notification of Reasons for Refusal mailed Mar. 15, 2022”, w/ English claims, 12 pgs. cited by applicant

“Korean Application Serial No. 10-2016-0057390, Notice of Preliminary Rejection/Reasons for Refusal mailed Nov. 14, 2022”, w/ English Translation, 6 pgs. cited by applicant

“Korean Application Serial No. 10-2016-0057390, Response filed Jul. 28, 2023 to Notice of Preliminary Rejection mailed May 28, 2023”, w English claims, 6 pgs. cited by applicant

Korean Application Serial No. 10-2024-0025363, Notice of Preliminary Rejection mailed Oct. 30, 2024, W/English Translation, 6 pgs. cited by applicant

“Korean Application Serial No. 10-2024-0025363, Response filed Dec. 30, 2024 to Notice of Preliminary Rejection mailed Oct. 30, 2024”, w English claims, 4 pgs. cited by applicant

---

*Primary Examiner:* Bennett; Charlee J. C.

*Attorney, Agent or Firm:* Schwegman Lundberg & Woessner, P.A.

---

## **Background/Summary**

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional patent application of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/863,835, filed on Apr. 30, 2020, which is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 15/662,553, filed on Jul. 28, 2017, now U.S. Pat. No. 10,655,225, issued on May 19, 2020, which is a divisional application of U.S. patent application Ser. No. 14/710,132, filed May 12, 2015 entitled “Substrate Pedestal Module Including Backside Gas Delivery Tube and Method of Making”, now U.S. Pat. No. 9,738,975, issued on Aug. 22, 2017, the content of each of which is incorporated herein in its entirety.

## **FIELD OF THE INVENTION**

(1) This invention pertains to semiconductor substrate processing apparatuses for processing semiconductor substrates, and may find particular use in plasma-enhanced chemical vapor depositions processing apparatuses operable to deposit thin films.

## **BACKGROUND**

(2) Semiconductor substrate processing apparatuses are used to process semiconductor substrates by techniques including etching, physical vapor deposition (PVD), chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), atomic layer deposition (ALD), plasma-enhanced atomic layer deposition (PEALD), pulsed deposition layer (PDL), plasma-enhanced pulsed deposition layer (PEPDL), and resist removal. One type of semiconductor substrate processing apparatus is a plasma processing apparatus that includes a reaction chamber containing upper and lower electrodes wherein a radio frequency (RF) power is applied between the electrodes to excite a process gas into plasma for processing semiconductor substrates in the

reaction chamber.

## SUMMARY

(3) Disclosed herein is a semiconductor substrate processing apparatus for processing semiconductor substrates. The semiconductor substrate processing apparatus comprises a vacuum chamber including a processing zone in which a semiconductor substrate may be processed, a process gas source in fluid communication with the vacuum chamber for supplying a process gas into the vacuum chamber, a showerhead module through which process gas from the process gas source is supplied to the processing zone of the vacuum chamber, and a substrate pedestal module. The substrate pedestal module includes a platen made of ceramic material having an upper surface configured to support a semiconductor substrate thereon during processing, a stem made of ceramic material having an upper stem flange that supports the platen, and a backside gas tube made of ceramic material that is located in an interior of the stem. The backside gas tube includes an upper gas tube flange that is located between a lower surface of the platen and an upper surface of the upper stem flange wherein the backside gas tube is in fluid communication with at least one backside gas passage of the platen and the backside gas tube is configured to supply a backside gas to a region below a lower surface of a semiconductor substrate that is to be supported on the upper surface of the platen during processing.

(4) Also disclosed herein is a substrate pedestal module of a semiconductor substrate processing apparatus. The substrate pedestal module comprises a platen having an upper surface configured to support a semiconductor substrate thereon during processing, a stem having an upper stem flange that supports the platen, and a backside gas tube that is located in an interior of the stem. The backside gas tube includes an upper gas tube flange that is located between a lower surface of the platen and an upper surface of the upper stem flange wherein the backside gas tube is in fluid communication with at least one backside gas passage of the platen and the backside gas tube is configured to supply a backside gas to a region below a lower surface of a semiconductor substrate that is to be supported on the upper surface of the platen during processing.

(5) Further disclosed herein is a method of making a substrate pedestal module of a semiconductor substrate processing apparatus. The method of making the substrate pedestal module includes positioning an upper ceramic surface of an upper gas tube flange against a lower ceramic surface of a platen, and diffusion bonding the upper ceramic surface of the upper gas tube flange to the lower ceramic surface of the platen so as to form a vacuum seal.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) FIG. 1 illustrates a schematic diagram showing an overview of a chemical deposition apparatus in accordance with embodiments disclosed herein.

(2) FIG. 2 shows a cross section of a substrate pedestal module in accordance with an embodiment as disclosed herein.

(3) FIG. 3 illustrates a cross section of a substrate pedestal module in accordance with an embodiment as disclosed herein.

### DETAILED DESCRIPTION

(4) In the following detailed description, numerous specific embodiments are set forth in order to provide a thorough understanding of the apparatus and methods disclosed herein. However, as will be apparent to those skilled in the art, the present embodiments may be practiced without these specific details or by using alternate elements or processes. In other instances, well-known processes, procedures, and/or components have not been described in detail so as not to unnecessarily obscure aspects of embodiments disclosed herein. As used herein the term “about” refers to  $\pm 10\%$ .

(5) As indicated, present embodiments provide apparatus and associated methods for processing a semiconductor substrate in a semiconductor substrate processing apparatus such as a chemical vapor deposition apparatus or a plasma-enhanced chemical vapor deposition apparatus. The apparatus and methods are particularly applicable for use in conjunction with high temperature processing of semiconductor substrates such as a high temperature deposition processes wherein a semiconductor substrate being processed is heated to temperatures greater than about 550° C., such as about 550° C. to about 650° C. or more.

(6) Embodiments disclosed herein are preferably implemented in a plasma-enhanced chemical deposition apparatus (i.e. PECVD apparatus, PEALD apparatus, or PEPDL apparatus), however, they are not so limited. FIG. 1 provides a simple block diagram depicting various semiconductor substrate plasma processing apparatus components arranged for implementing embodiments as disclosed herein. As shown, a semiconductor substrate plasma processing apparatus **300** includes a vacuum chamber **324** that serves to contain plasma in a processing zone, which is generated by a capacitor type system including a showerhead module **314** having an upper RF electrode (not shown) therein working in conjunction with a substrate pedestal module **320** having a lower RF electrode (not shown) therein. At least one RF generator is operable to supply RF energy into a processing zone above an upper surface of a semiconductor substrate **316** in the vacuum chamber **324** to energize process gas supplied into the processing zone of the vacuum chamber **324** into plasma such that a plasma deposition process may be performed in the vacuum chamber **324**. For example, a high-frequency RF generator **302** and a low-frequency RF generator **304** may each be connected to a matching network **306**, which is connected to the upper RF electrode of the showerhead module **314** such that RF energy may be supplied to the processing zone above the semiconductor substrate **316** in the vacuum chamber **324**.

(7) The power and frequency of RF energy supplied by matching network **306** to the interior of the vacuum chamber **324** is sufficient to generate plasma from the process gas. In an embodiment both the high-frequency RF generator **302** and the low-frequency RF generator **304** are used, and in an alternate embodiment, just the high-frequency RF generator **302** is used. In a process, the high-frequency RF generator **302** may be operated at frequencies of about 2-100 MHz; in a preferred embodiment at 13.56 MHz or 27 MHz. The low-frequency RF generator **304** may be operated at about 50 kHz to 2 MHz; in a preferred embodiment at about 350 to 600 kHz. The process parameters may be scaled based on the chamber volume, substrate size, and other factors. Similarly, the flow rates of process gas, may depend on the free volume of the vacuum chamber or processing zone.

(8) An upper surface of the substrate pedestal module **320** supports a semiconductor substrate **316** during processing within the vacuum chamber **324**. The substrate pedestal module **320** can include a chuck to hold the semiconductor substrate and/or lift pins to raise and lower the semiconductor substrate before, during and/or after the deposition and/or plasma treatment processes. In an alternate embodiment, the substrate pedestal module **320** can include a carrier ring to raise and lower the semiconductor substrate before, during and/or after the deposition and/or plasma treatment processes. The chuck may be an electrostatic chuck, a mechanical chuck, or various other types of chuck as are available for use in the industry and/or research. Details of a lift pin assembly for a substrate pedestal module including an electrostatic chuck can be found in commonly-assigned U.S. Pat. No. 8,840,754, which is incorporated herein by reference in its entirety. Details of a carrier ring for a substrate pedestal module can be found in commonly-assigned U.S. Pat. No. 6,860,965, which is incorporated herein by reference in its entirety. A backside gas supply **341** is operable to supply a heat transfer gas or purge gas through the substrate pedestal module **320** to a region below a lower surface of the semiconductor substrate during processing. The substrate pedestal module **320** includes the lower RF electrode therein wherein the lower RF electrode is preferably grounded during processing, however in an alternate embodiment, the lower RF electrode may be supplied with RF energy during processing.



(9) To process a semiconductor substrate in the vacuum chamber **324** of the semiconductor substrate plasma processing apparatus **300**, process gases are introduced from a process gas source **362** into the vacuum chamber **324** via inlet **312** and showerhead module **314** wherein the process gas is formed into plasma with RF energy such that a film may be deposited onto the upper surface of the semiconductor substrate. In an embodiment, multiple source gas lines **310** may be connected to a heated manifold **308**. The gases may be premixed or supplied separately to the chamber. Appropriate valving and mass flow control mechanisms are employed to ensure that the correct gases are delivered through the showerhead module **314** during semiconductor substrate processing. During the processing, a backside heat transfer gas or purge gas is supplied to a region below a lower surface of the semiconductor substrate supported on the substrate pedestal module **320**. Preferably, the processing is at least one of chemical vapor deposition processing, plasma-enhanced chemical vapor deposition processing, atomic layer deposition processing, plasma-enhanced atomic layer deposition processing, pulsed deposition layer processing, or plasma-enhanced pulsed deposition layer processing.

(10) In certain embodiments, a system controller **162** is employed to control process conditions during deposition, post deposition treatments, and/or other process operations. The controller **162** will typically include one or more memory devices and one or more processors. The processor may include a CPU or computer, analog and/or digital input/output connections, stepper motor controller boards, etc.

(11) In certain embodiments, the controller **162** controls all of the activities of the apparatus. The system controller **162** executes system control software including sets of instructions for controlling the timing of the processing operations, frequency and power of operations of the low-frequency RF generator **304** and the high-frequency RF generator **302**, flow rates and temperatures of precursors and inert gases and their relative mixing, temperature of a semiconductor substrate **316** supported on an upper surface of the substrate pedestal module **320** and a plasma exposed surface of the showerhead module **314**, pressure of the vacuum chamber **324**, and other parameters of a particular process. Other computer programs stored on memory devices associated with the controller may be employed in some embodiments.

(12) Typically there will be a user interface associated with controller **162**. The user interface may include a display screen, graphical software displays of the apparatus and/or process conditions, and user input devices such as pointing devices, keyboards, touch screens, microphones, etc.

(13) A non-transitory computer machine-readable medium can comprise program instructions for control of the apparatus. The computer program code for controlling the processing operations can be written in any conventional computer readable programming language: for example, assembly language, C, C++, Pascal, Fortran or others. Compiled object code or script is executed by the processor to perform the tasks identified in the program.

(14) The controller parameters relate to process conditions such as, for example, timing of the processing steps, flow rates and temperatures of precursors and inert gases, temperature of the semiconductor substrate, pressure of the chamber and other parameters of a particular process. These parameters are provided to the user in the form of a recipe, and may be entered utilizing the user interface.

(15) Signals for monitoring the process may be provided by analog and/or digital input connections of the system controller. The signals for controlling the process are output on the analog and digital output connections of the apparatus.

(16) The system software may be designed or configured in many different ways. For example, various chamber component subroutines or control objects may be written to control operation of the chamber components necessary to carry out deposition processes. Examples of programs or sections of programs for this purpose include substrate timing of the processing steps code, flow rates and temperatures of precursors and inert gases code, and a code for pressure of the vacuum chamber **324**.

(17) FIGS. 2 and 3 illustrate cross sections of a substrate pedestal module 320 according to embodiments as disclosed herein. As shown in FIGS. 2 and 3, the substrate pedestal module 320 includes a platen 205 made of ceramic material having an upper surface 206 that supports a semiconductor substrate thereon during processing of the semiconductor substrate. A stem 210 made of ceramic material extends downwardly from a lower surface 208 of the platen 205 wherein the stem 210 includes an upper stem flange 215 that supports the platen 205 and a lower stem flange 220 that may be attached to a base or adapter (not shown) such that the substrate pedestal module 320 may be supported in a vacuum chamber of a semiconductor substrate processing apparatus. By making the substrate pedestal module 320 from ceramic materials rather than a metal material, such as aluminum or an aluminum alloy, the substrate pedestal module 320 may withstand high temperatures achieved during high temperature substrate processing, such as temperatures greater than about 550° C. or temperatures greater than about 650° C.

(18) The substrate pedestal module 320 includes a backside gas tube 250 made of ceramic that is located in an interior 213 of the stem 210. In an embodiment, the backside gas tube 250 is centrally located in the interior 213 of the stem 210. In an alternate embodiment, the backside gas tube 250 is located off-center in the interior of the stem 210. The backside gas tube 250 includes an upper gas tube flange 253 that is located between a lower surface 208 of the platen 205 and an upper surface of the upper stem flange 215. The backside gas tube 250 is in fluid communication with at least one backside gas passage 207 of the platen 205 such that a backside gas may be supplied through the backside gas tube 250 and the backside gas passage 207 to a region below a lower surface of a semiconductor substrate that is supported on the upper surface 206 of the platen 205 during processing. In an embodiment, an outer diameter of the upper stem flange 215 and an outer diameter of the upper gas tube flange 253 can be equal to or about equal to an outer diameter of the platen 205. Preferably, the outer diameter of the upper stem flange 215 and an outer diameter of the upper gas tube flange 253 are less than the outer diameter of the platen 205. The outer diameter of the upper stem flange 215, the outer diameter of the upper gas tube flange 253 and the outer diameter of the stem 210, as well as the thickness of the stem 210 wall are selected such that the interior 213 of the stem 210, which is maintained at atmospheric pressure during processing, can accommodate electrical connections that are disposed in the interior 213 thereof as well as withstand the pressure differential between the vacuum pressure surrounding the stem 210 and the atmospheric pressure in the interior 213 of the stem 210. Preferably, an outer periphery of the upper gas tube flange 253 is located on a radius no greater than ¼ the outer diameter of the platen 205.

(19) The platen 205 can include at least one electrostatic clamping electrode 209 embedded therein wherein the at least one electrostatic clamping electrode 209 is operable to electrostatically clamp a semiconductor substrate on the upper surface 206 of the platen 205 during processing. As shown in FIG. 2 the platen 205 can also include a lower RF electrode 265 that may be grounded or supplied with RF power during processing of a semiconductor substrate. Preferably, as shown in FIG. 3, the platen 205 includes only a single electrode 209a embedded therein that serves as both an electrostatic clamping electrode and a RF electrode. Referring back to FIGS. 2 and 3, the platen 205 can also include at least one heater 260 embedded therein that is operable to control the temperature across the upper surface 206 of the platen 205 and thereby the temperature across the semiconductor substrate during processing. The at least one heater 260 can include electrically resistive heater film and/or one or more thermoelectric modules. Preferably, electrical connections to the at least one electrostatic clamping electrode 209, the at least one heater 260, the single electrode 209a, and/or the lower RF electrode 265 that are disposed in the interior 213 of the stem 210 extend through respective openings 251 in the upper gas tube flange 253 such that the electrical connections may be respectively connected to electrical contacts (not shown) formed in the platen 205 that are in electrical communication with the respective at least one electrostatic clamping electrode 209, the at least one heater 260, the single electrode 209a and/or the lower RF electrode 265. In this manner, the at least one electrostatic clamping electrode 209, the at least one

heater **260**, the single electrode **209a**, and/or the lower RF electrode **265** may be powered during processing of a semiconductor substrate.

(20) In an embodiment, the platen **205** can include discrete layers that are diffusion bonded together wherein the at least one electrostatic clamping electrode **209**, the lower RF electrode **265** (or the single electrode **209a**), and the at least one heater **260** can be sandwiched between the discrete layers of the platen **205**. The upper surface **206** of the platen **205** preferably includes a mesa pattern **206a** formed therein wherein a lower surface of a semiconductor substrate is supported on the mesa pattern **206a** and a backside purge gas or a backside heat transfer gas can be supplied to the region below the semiconductor substrate between the mesas of the mesa pattern **206a**. An exemplary embodiment of a mesa pattern and method of forming a mesa pattern can be found in commonly assigned U.S. Pat. No. 7,869,184, which is hereby incorporated herein in its entirety.

(21) The exposed surfaces of the platen **205**, the stem **210**, and the backside gas tube **250** are made of ceramic material which preferably does not lead to substrate contamination during processing when the platen **205**, stem **210**, and/or backside gas tube **250** are exposed to processing conditions. Preferably, exposed surfaces of the platen **205**, the stem **210**, and/or the backside gas tube **250** are formed of aluminum nitride. The upper surface of the upper gas tube flange **253** is preferably diffusion bonded to the lower surface **208** of the platen **205** and the lower surface of the upper gas tube flange **253** is preferably diffusion bonded to an upper surface of the upper stem flange **215** so as to form a vacuum seal between the interior **213** of the stem **210** and a vacuum chamber in which the substrate pedestal module **320** is disposed. Preferably, the contact area between the upper surface of the upper gas tube flange **253** and the lower surface **208** of the platen **205** is about equal to the contact area between the lower surface of the upper gas tube flange **253** and the upper surface of the upper stem flange **215**. More preferably, the contact area between the upper surface of the upper gas tube flange **253** and the lower surface **208** of the platen **205** is equal to the contact area between the lower surface of the upper gas tube flange **253** and the upper surface of the upper stem flange **215**. FIG. 2 illustrates an embodiment of the substrate pedestal module **320** wherein the contact area between the upper surface of the upper gas tube flange **253** and the lower surface **208** of the platen **205** is equal to the contact area between the lower surface of the upper gas tube flange **253** and the upper surface of the upper stem flange **215**. In this embodiment, a gap **230** is formed between a portion of the upper gas tube flange **253** and the lower surface **208** of the platen **205**. In an alternate embodiment, as illustrated in FIG. 3, the substrate pedestal module **320** can be formed such that there is no gap between the upper surface of the upper gas tube flange **253** and the lower surface **208** of the platen **205**.

(22) Further disclosed herein is a method of making a substrate pedestal module **320**. The method of making the substrate pedestal module **320** includes positioning an upper ceramic surface of an upper gas tube flange **253** against a lower ceramic surface **208** of a platen **205**. The upper ceramic surface of the upper gas tube flange **253** is diffusion bonded to the lower ceramic surface **208** of the platen **205**. In an embodiment, an upper ceramic surface of an upper stem flange **215** is simultaneously positioned against a lower ceramic surface of the upper gas tube flange **253** wherein the upper ceramic surface of the upper gas tube flange **253** is simultaneously diffusion bonded to the lower ceramic surface of the platen **205** while the upper ceramic surface of the upper stem flange **215** is diffusion bonded to the lower ceramic surface of the upper gas tube flange **253**. To simultaneously diffusion bond the upper ceramic surface of an upper stem flange **215** to the lower ceramic surface of the upper gas tube flange **253** and the upper ceramic surface of the upper gas tube flange **253** to the lower ceramic surface of the platen **205** the contact area between the upper ceramic surface of the upper stem flange **215** and the lower ceramic surface of the upper gas tube flange **253** as well as the contact area between the upper ceramic surface of the upper gas tube flange **253** and the lower ceramic surface **208** of the platen **205** are equal or about equal such that the ceramic surfaces that are diffusion bonded together form a vacuum seal. In this manner, the

platen 205, the stem 210, and the backside gas tube 250 can be integrally connected in a single diffusion bonding process.

(23) In an alternate embodiment, if the contact area between the upper ceramic surface of the upper gas tube flange 253 and the lower ceramic surface 208 of the platen 205 is not equal to the contact area between the lower ceramic surface of the upper gas tube flange 253 and the upper ceramic surface of the upper stem flange 215 then the upper ceramic surface of the upper gas tube flange 253 is first diffusion bonded to the lower ceramic surface 208 of the platen 205 wherein the upper ceramic surface of the upper stem flange 215 is subsequently bonded to a lower ceramic surface of the upper gas tube flange 253.

(24) While the substrate pedestal module of the semiconductor substrate processing apparatus has been described in detail with reference to specific embodiments thereof, it will be apparent to those skilled in the art that various changes and modifications can be made, and equivalents employed, without departing from the scope of the appended claims.

## Claims

1. A method of processing a substrate, the method comprising: placing the substrate on a substrate pedestal, the substrate pedestal including a platen formed from a ceramic material having an upper surface configured to support the substrate during processing, a stem of the substrate pedestal being formed from a ceramic material and having an upper-stem flange to support the platen, a backside-gas tube is formed from a ceramic material that is located in an interior portion of the stem, the backside-gas tube including an upper gas-tube flange located between a lower surface of the platen and an upper surface of the upper-stem flange, the backside-gas tube being in fluid communication with at least one backside gas passage of the platen, the backside-gas tube being configured to supply a backside gas to a region below a lower surface of the substrate that is to be supported on an upper surface of the platen during processing; forming a vacuum seal through the upper surface of the upper gas-tube flange to the lower surface of the platen through diffusion bonding; and supplying a gas selected from gases including a backside heat-transfer gas and a purge gas through the backside-gas tube to a region below and in fluid communication with at least a central portion of a lower surface of the substrate supported on the upper surface of the platen during processing.
2. The method of claim 1, wherein the substrate pedestal is located within a vacuum chamber with a showerhead located in the vacuum chamber and above the substrate pedestal, the method further comprising: supplying a process gas through the showerhead into the vacuum chamber, the process gas to be supplied from a process gas source that is in fluid communication with the vacuum chamber; and supplying the process gas from the showerhead into a processing zone located between the showerhead and the substrate located on the substrate pedestal.
3. The method of claim 1, wherein the processing comprises at least one type of deposition process selected from deposition processes including chemical vapor deposition, plasma-enhanced chemical vapor deposition, atomic layer deposition, plasma-enhanced atomic layer deposition, pulsed deposition layer, and plasma-enhanced pulsed deposition layer.
4. The method of claim 1, further comprising placing the substrate on a mesa pattern formed on the upper surface of the platen, the mesa pattern to provide gas passages between mesas of the mesa pattern and a lower surface of the substrate supported on the mesa pattern.
5. The method of claim 1, wherein, the backside-gas tube includes at least one opening in the upper gas-tube flange such that at least one respective electrical connection can extend through the upper gas-tube flange.
6. A method of processing a substrate, the method comprising: placing the substrate on a platen of a substrate pedestal, the substrate pedestal including a stem coupled to the platen; injecting at least one backside gas into a backside-gas tube that is located in an interior portion of the stem, the backside-gas tube including an upper gas-tube flange, the backside-gas tube being in fluid

communication with at least one backside gas passage of the platen; forming a vacuum seal through the upper surface of the upper gas-tube flange to the lower surface of the platen through diffusion bonding; and supplying a backside gas, the backside gas being selected from gases including a backside heat-transfer gas and a purge gas through the backside-gas tube to a region below and in fluid communication with at least a central portion of a lower surface of the substrate supported on an upper surface of the platen during processing.

7. The method of claim 6, wherein the backside-gas tube includes at least one opening in the upper gas-tube flange such that at least one respective electrical connection can extend through the upper gas-tube flange.

8. The method of claim 6, wherein: the platen is formed from a ceramic material; and the stem of the substrate pedestal is formed from a ceramic material.

9. The method of claim 6, wherein the backside-gas tube is formed from a ceramic material.

10. The method of claim 6, wherein the backside-gas tube includes the upper gas-tube flange that is located between a lower surface of the platen and an upper surface of the upper-stem flange.

11. The method of claim 6, wherein the substrate pedestal is to be located within a vacuum chamber with a showerhead located in the vacuum chamber and above the substrate pedestal, the method further comprising: supplying a process gas through the showerhead into the vacuum chamber, the process gas to be supplied from a process gas source that is in fluid communication with the vacuum chamber; and supplying the process gas from the showerhead into a processing zone located between the showerhead and the substrate located on the substrate pedestal.

12. The method of claim 6, further comprising placing the substrate on a mesa pattern formed on an upper surface of the platen, the mesa pattern to provide gas passages between mesas of the mesa pattern and a lower surface of the substrate supported on the mesa pattern.

13. A method of forming a substrate pedestal, the method comprising: forming a platen having an upper surface configured to support a substrate during processing on the substrate pedestal; forming a stem of the substrate pedestal to have an upper-stem flange to support the platen; forming a backside-gas tube within an interior portion of the stem, the backside-gas tube including an upper gas-tube flange located between a lower surface of the platen and an upper surface of the upper-stem flange, the backside-gas tube to be in fluid communication with at least one backside gas passage of the platen, the backside-gas tube being configured to supply a backside gas to a region below and in fluid communication with at least a central portion of a lower surface of the substrate that is to be supported on the upper surface of the platen during processing; and diffusion bonding an upper surface of the upper gas-tube flange to the lower surface of the platen so as to form a vacuum seal.

14. The method of claim 13, further comprising forming each of the platen, the stem, and the backside-gas tube from a ceramic material.

15. The method of claim 13, wherein the plate, the stem, and the backside-gas tube are formed from aluminum nitride.

16. The method of claim 13, further comprising forming a mesa pattern formed on the upper surface of the platen, the mesa pattern to provide gas passages between mesas of the mesa pattern and a lower surface of the substrate to be supported on the mesa pattern.

17. The method of claim 13, further comprising forming at least one electrical connection to extend through the upper gas-tube flange.

---